

APO-MELOXICAM

Meloxicam Tablets 7.5 mg

Anti-inflammatory Analgesic Agent

Pharmacology

Meloxicam is a nonsteroidal anti-inflammatory drug (NSAID) that exhibits anti-inflammatory, analgesic, and antipyretic properties in animals. Meloxicam showed potent anti-inflammatory activity in all standard models of inflammation. NSAIDs are believed to exert their pharmacologic effects primarily through inhibition of the enzyme of the cyclooxygenase (COX). In turn, inhibition of this enzyme leads to an inhibition of biosynthesis of prostaglandins and other autacoids, substances which are potent biological mediators involved in diverse physiologic functions as well as pathologic conditions.

Meloxicam has been shown to inhibit COX-2 in several *in vitro* and *ex vivo* test systems. The inhibition of thromboxane in platelets, and consequently platelet aggregation, occurs via inhibition of COX-1. Meloxicam inhibition of thromboxane in platelets (via COX-1) is dose dependent and incomplete at anti-inflammatory doses. No significant inhibition of platelet aggregation has been observed with meloxicam at the recommended therapeutic doses of 7.5 and 15 mg once daily.

Inhibition of COX-2 also inhibits the production of systemic prostacyclin. Inhibition of prostacyclin may have a pro-thrombotic effect.

Pharmacokinetics

Absorption: The absolute bioavailability of meloxicam capsules was 89% following a single oral dose of 30 mg compared with 30 mg IV bolus injection. Meloxicam capsules have been shown to be bioequivalent to meloxicam tablets. After multiple oral doses the pharmacokinetics of meloxicam capsules were dose-proportional over the range of 7.5 mg to 15 mg. Mean C_{max} was achieved within four to five hours after a 7.5 mg meloxicam tablet was taken under fasted conditions, indicating a prolonged drug absorption. The rate or extent of absorption was not affected by multiple dose administration, suggesting linear pharmacokinetics. With multiple dosing, steady state conditions were reached by day 5.

Distribution: The mean volume of distribution (V_d) of meloxicam is approximately 10 L. Meloxicam is ~99.4% bound to human plasma proteins (primarily albumin) within the therapeutic dose range. Meloxicam concentrations in synovial fluid, after a single oral dose, range from 40% to 50% of those in plasma. The free fraction in synovial fluid is 2.5 times higher than in plasma, due to the lower albumin content in synovial fluid as compared to plasma. The significance of this penetration is unknown.

Metabolism: Meloxicam is almost completely metabolized to four pharmacologically inactive metabolites. The major metabolites, 5'-carboxy meloxicam (60% of dose), from P-450 mediated metabolism was formed by oxidation of an intermediate metabolite 5'-hydroxymethyl meloxicam which is also excreted to a lesser extent (9% of dose).

Excretion: Meloxicam excretion is predominantly in the form of metabolites, and occurs to equal extents in the urine and feces. Only traces of the unchanged parent compound are excreted in the urine and feces. The mean elimination half-life ($t_{1/2}$) ranges from 15 hours to 20 hours.

Indication

APO-MELOXICAM (meloxicam) tablets are indicated for symptomatic treatment of:

- rheumatoid arthritis
- painful osteoarthritis (arthrosis, degenerative joint disease) in adults
- Ankylosing Spondylitis

Recommended Dose

Osteoarthritis: 7.5 mg once daily. If necessary, the dose may be increased to 15 mg once daily.

Rheumatoid arthritis: 15 mg once daily. According to the therapeutic response, the dose may be reduced to 7.5 mg once daily.

In patients with increased risks of adverse reactions, treatment should be started at the dose of 7.5 mg once daily. In dialysis patients with severe renal failure, the dose should not exceed 7.5 mg/day.

The maximum recommended daily dose of APO-MELOXICAM (meloxicam) tablets is 15 mg.

As a dosage for use in children has yet to be established, usage should be restricted to adults.

APO-MELOXICAM may be taken without regard to timing of meals.

Ankylosing spondylitis: 15mg once daily. According to the therapeutic response, the dose may be reduced to 7.5 mg once daily.

Mode of Administration: Oral route.

Contraindication

Known or suspected hypersensitivity to meloxicam or any other component of APO-MELOXICAM (meloxicam). APO-MELOXICAM should not be used in patients in whom acute asthmatic attacks or symptoms of asthma, urticaria, nasal polyps, anaphylaxis, rhinitis, angioedema or other allergic manifestations are precipitated by ASA or other nonsteroidal anti-inflammatory agents since cross-sensitivity may exist. Fatal anaphylactoid reactions have occurred in such individuals treated with other NSAIDs.

APO-MELOXICAM is also contraindicated in: active peptic ulceration; severe hepatic insufficiency; non-dialysed severe renal insufficiency; children and adolescents aged less than 15 years; pregnancy or breastfeeding.

APO-MELOXICAM is not recommended for use with other NSAIDs because of the absence of any evidence demonstrating synergistic benefits and the potential for additive side effects.

Warnings

Risk of GI Ulceration, bleeding and perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Precautions

Peptic ulceration, perforation and gastrointestinal bleeding, sometimes severe, and occasionally fatal, can occur at any time, with or without symptoms in patients treated with nonsteroidal anti-inflammatory drugs (NSAIDs). Minor upper GI problems, such as dyspepsia, are common, and may also occur at any time during NSAID therapy. Therefore, physicians should remain alert for ulceration and bleeding in patients treated with non-steroidal anti-inflammatory drugs, even in the absence of previous GI tract symptoms. Patients should be informed about the signs and/or symptoms of serious GI toxicity and the steps to take if they occur.

NSAIDs should be prescribed with extreme caution in patients with a prior history of ulcer disease or gastrointestinal bleeding. Most spontaneous reports of fatal GI events are in elderly or debilitated patients and therefore special care should be taken in treating this population. To minimize the potential risk for an adverse GI event, the lowest effective dose should be used for the shortest possible duration. For high risk patients, alternate therapies that do not involve NSAIDs should be considered.

Cross Sensitivity: Patients sensitive to any one of the non-steroidal anti-inflammatory drugs may be sensitive to any of the other NSAIDs also.

Use in Children: safety and effectiveness of APO-MELOXICAM in pediatric patients below the age of 18 years have not been evaluated.

Renal Function: Caution should be used when initiating treatment with meloxicam in patients with considerable dehydration. It is advisable to rehydrate patients first and then start therapy with meloxicam. Caution is also recommended in patients with pre-existing kidney disease. No dose reduction is required in patients with mild or moderate renal impairment (i.e. in patients with a creatinine clearance of greater than 15 ml/min or 0.25 ml/sec).

The extent to which metabolites may accumulate in patients with renal failure has not been studied with meloxicam. As with other NSAIDs, metabolites of which are excreted by the kidney, patients with significantly impaired renal function should be more closely monitored.

Drug Interactions

Ace Inhibitors: Reports suggest that NSAIDs may diminish the antihypertensive effect of angiotensin-converting enzyme (ACE) inhibitors. This interaction should be given consideration in patients taking NSAIDs concomitantly with ACE inhibitors.

Aspirin: As with other NSAIDs, concomitant administration of meloxicam and aspirin is not generally recommended because of the potential for increased adverse effects. Concomitant administration of low-dose aspirin with APO-MELOXICAM may result in an increased rate of GI ulceration or other complications, compared to use of meloxicam alone.

Cholestyramine: Cholestyramine binds meloxicam in the gastrointestinal tract to a faster elimination of meloxicam.

Diuretics: Treatment with NSAIDs is associated with the potentials for acute renal insufficiency in patients who are dehydrated. Patients who are receiving APO-MELOXICAM and diuretics at the same time should be sufficiently hydrated and be monitored for renal function prior to initiating treatment.

Lithium: It is reported that NSAIDs increases lithium plasma levels. Therefore, it is recommended that plasma lithium levels be monitored when initiating, adjusting and discontinuing APO-MELOXICAM.

Methotrexate: Like any other NSAIDs, APO-MELOXICAM may increase the haemologic toxicity of methotrexate. In such situation, strict monitoring of blood cell count is recommended.

Anticoagulants: APO-MELOXICAM will increase the risk of bleeding. Should such co-prescribing can not be avoided, intense monitoring of the effects of anticoagulants is needed.

Cyclosporin: Nephrotoxicity of cyclosporin may be enhanced by NSAIDs via renal prostaglandin-mediated effects. During combined treatment, renal function is to be measured.

No relevant pharmacokinetic drug-drug interactions were detected with respect to the concomitant administration of antacids, cimetidine, digoxin, and furosemide.

Use in Pregnancy

There are no studies with APO-MELOXICAM in pregnant women. In late pregnancy, as with other NSAIDs, APO-MELOXICAM should be avoided because it may cause premature closure of the ductus arteriosus. APO-MELOXICAM should be used during the first two trimesters of pregnancy only if the potential benefit justifies the potential risk to the fetus.

Use in Nursing Mothers

Studies of meloxicam excretion in human milk have not been conducted; however, meloxicam was excreted in the milk of lactating rats at concentrations higher than those in plasma. Because of the potential for serious adverse reactions in nursing infants from APO-MELOXICAM, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

Adverse Effects

Adverse effects which may be causally related to the administration of APO-MELOXICAM are followed by a reference number. The incidence of those rare events is hard to quantify and all are assumed to occur with a frequency of < 0.1%.

Gastrointestinal: More common than 1%: Dyspepsia, nausea, vomiting, abdominal pain, constipation, flatulence and diarrhoea.

0.1-1%: Transient irregularity of liver function parameters (eg., raised transaminases or bilirubin), oesophagitis, eructation, gastroduodenal ulcer, occult or macroscopic gastrointestinal bleeding. Less common than 0.1%: Colitis, gastrointestinal perforation.

Haematological: More common than 1%: Anemia. 0.1 – 1%: Instability of blood count, inclusive of differential white cell count, leukopenia and thrombocytopenia. Concomitant administration of myelotoxic drug, in particular, methotrexate seems to be a inclining factor to the onset of cytopenia.

Dermatological: More common than 1%: Pruritus and skin rash. 0.1 – 1%: Stomatitis and urticaria. Less common than 0.1%: Photosensitisation.

On exceptional occasions, bullous reactions, Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis may develop.

Respiratory: Less common than 0.1%: There are reports about the onset of acute asthma in certain individuals following the administration of aspirin or other NSAIDs, including APO-MELOXICAM.

Central Nervous System: More common than 1%: Headache and light-headedness. 0.1 – 1%: Tinnitus, vertigo and drowsiness.

Cardiovascular: More common than 1%: Oedema. 0.1 – 1%: Increase of blood pressure, palpitations and flushes.

Genitourinary: 0.1 – 1%: Atypical renal function parameters (increased serum creatinine and/or serum urea).

Less common than 1%: Acute renal failure

Hypersensitivity Reactions: Less common than 0.1%. Immediate hypersensitivity reactions inclusive of anaphylactoid/anaphylactic reactions and angioedema

Symptoms and Treatment of Overdose

There is limited experience with meloxicam overdose. Four cases have taken 6 to 11 times the highest recommended dose; all recovered. Cholestyramine is known to accelerate the clearance of meloxicam.

Symptoms following acute NSAID overdose are usually limited to lethargy, drowsiness, nausea, vomiting, and epigastric pain, which are generally reversible with supportive care. Gastrointestinal bleeding can occur. Severe poisoning may result in hypertension, acute renal failure, hepatic dysfunction, respiratory depression, coma, convulsions, cardiovascular collapse, and cardiac arrest. Anaphylactoid reactions have been reported with therapeutic ingestion of NSAIDs, and may occur following an overdose.

Patients should be managed with symptomatic and supportive care following an NSAID overdose. In cases of acute overdose, gastric lavage followed by activated charcoal is recommended. Gastric lavage performed more than one hour after overdose has little benefit in the treatment of overdose. Administration of activated charcoal is recommended for patients who present 1 – 2 hours after overdose. For substantial overdose or severely symptomatic patients, activated charcoal may be administered repeatedly. Accelerated removal of meloxicam by 4gm oral doses of cholestyramine given three times a day was demonstrated in a clinical trial. Administration of cholestyramine may be useful following an overdose. Forced diuresis, alkalinization of urine, hemodialysis, or hemoperfusion may not be useful due to high protein binding.

Product Description/Availability of Dosage Forms

APO-MELOXICAM Tablets 7.5mg: Pastel yellow, round, biconvex tablet. Engraved "APO" on one side and "7.5" on the other. Available in PVC blisters of 500 tablets.

Storage Conditions: Store below 30°C in a dry place.

Manufacturer

Apotex Inc.,
150 Signet Drive, Toronto Ontario, Canada M9L 1T9.

Date of Revision of Package Insert

3rd Mar 2016